

Table 1: Incline motion – acceleration vs. hanging mass (a_- group, including the zero-crossing points)

Hanging mass m (g)	Acceleration a (cm s^{-2})
0.00	35.43 ± 0.11
1.54	31.043 ± 0.019
2.54	28.337 ± 0.032
3.54	25.810 ± 0.016
4.55	23.357 ± 0.025
5.53	20.657 ± 0.021
6.55	18.127 ± 0.014
7.55	15.740 ± 0.037
8.51	13.373 ± 0.032
9.47	10.747 ± 0.013
10.43	8.340 ± 0.037
11.55	5.427 ± 0.032
12.55	2.930 ± 0.048
13.55	0.423 ± 0.027
14.56	0.0000 ± 0.0029
15.54	0.0000 ± 0.0029

Table 2: Incline motion – acceleration vs. hanging mass (a_+ group)

Hanging mass m (g)	Acceleration a (cm s^{-2})
16.56	1.357 ± 0.021
17.56	3.773 ± 0.034
18.52	6.203 ± 0.037
19.48	8.543 ± 0.034
20.44	10.973 ± 0.025
21.57	13.830 ± 0.038
22.57	16.060 ± 0.011
23.57	18.627 ± 0.024
24.58	20.933 ± 0.014
25.56	23.167 ± 0.039
26.58	25.613 ± 0.029
27.58	27.920 ± 0.011
28.54	30.253 ± 0.024
29.50	32.383 ± 0.013
30.46	34.823 ± 0.019
40.55	58.127 ± 0.059
60.09	99.42 ± 0.32
80.11	137.72 ± 0.48
100.19	175.04 ± 0.42

Table 3: Linear regression of a_- (sign-flipped) against m

Quantity	Value
Slope ($\text{cm s}^{-2} \text{g}^{-1}$)	2.5463 ± 0.0018
y-intercept (cm s^{-2})	-34.862 ± 0.014
x-intercept m_- (g)	13.6915 ± 0.0054

Table 4: Linear regression of a_+ against m (first 15 points only, to avoid the large- m nonlinearity)

Quantity	Value
Slope ($\text{cm s}^{-2} \text{g}^{-1}$)	2.3868 ± 0.0013
y-intercept (cm s^{-2})	-37.881 ± 0.031
x-intercept m_+ (g)	15.8708 ± 0.0051

Table 5: Comparison of the cart mass, friction force, and incline angle – incline-motion method vs. an independent method

Quantity	Incline-motion method	Other method	Error
Cart mass (g)	385.14 ± 0.27	382.70	0.64%
Friction force (N)	0.010686 ± 0.000036	0.011912 ± 0.000066	-10.29%
Incline angle ($^\circ$)	2.199	2.289	-3.91%

Table 6: Nonlinear curve fit of a_- against m

Quantity	Value
M (g)	384.86 ± 0.28
F (dyne)	$(1.5608 \pm 0.0063) \times 10^3$

Table 7: Nonlinear curve fit of a_+ against m

Quantity	Value
M (g)	377.70 ± 0.19
F (dyne)	911 ± 12

Table 8: Comparison of the cart mass and friction force – curve-fit method vs. an independent method

Quantity	Curve-fit method	Other method	Error
Cart mass (g)	381.28 ± 0.17	382.70	-0.37%
Friction force (N)	0.012358 ± 0.000067	0.011912 ± 0.000066	3.74%

Table 9: Raw measurements – incline motion, a_- group (3 trials per hanging mass, comma-separated)

Hanging mass m (g)	Acceleration readings a (cm s^{-2})
0.00	35.60,35.46,35.24
1.54	31.03,31.08,31.02
2.54	28.40,28.30,28.31
3.54	25.82,25.78,25.83
4.55	23.31,23.39,23.37
5.53	20.69,20.66,20.62
6.55	18.14,18.10,18.14
7.55	15.69,15.81,15.72
8.51	13.32,13.43,13.37
9.47	10.73,10.74,10.77
10.43	8.27,8.39,8.36
11.55	5.49,5.40,5.39
12.55	2.91,3.02,2.86
13.55	0.38,0.47,0.42
14.56	0,0,0
15.54	0,0,0

Table 10: Raw measurements – incline motion, a_+ group (3 trials per hanging mass, comma-separated)

Hanging mass m (g)	Acceleration readings a (cm s^{-2})
16.56	1.32,1.39,1.36
17.56	3.84,3.75,3.73
18.52	6.24,6.24,6.13
19.48	8.50,8.52,8.61
20.44	11.02,10.96,10.94
21.57	13.77,13.90,13.82
22.57	16.05,16.05,16.08
23.57	18.59,18.67,18.62
24.58	20.92,20.96,20.92
25.56	23.15,23.11,23.24
26.58	25.67,25.58,25.59
27.58	27.91,27.94,27.91
28.54	30.21,30.29,30.26
29.50	32.40,32.36,32.39
30.46	34.80,34.86,34.81
40.55	58.19,58.01,58.18
60.09	98.86,99.96,99.43
80.11	137.40,137.09,138.66
100.19	175.87,174.72,174.54

Table 11: Raw measurements – cart sliding left/right on horizontal track

Cart moving left (cm s ⁻²)	Cart moving right (cm s ⁻²)
-3.17	-3.21
-3.10	-3.05
-3.15	-3.13
-3.10	-3.19
-3.10	-3.24
-3.15	-3.09
-3.15	-2.98
-3.09	-2.97
-3.19	-2.97
-3.12	-3.10